

5.1 Amazon Q Introduction

Course objectives

By the end of this course, you will be able to:

- Identify the purpose of Amazon Q.
- Identify common uses for Amazon Q.
- Recognize possible benefits of using Amazon Q.

Amazon Q

Introduction

The Amazon Q family of products contains different services, all focused to help you streamline tasks, speed decision-making, and help spark creativity and innovation at work.

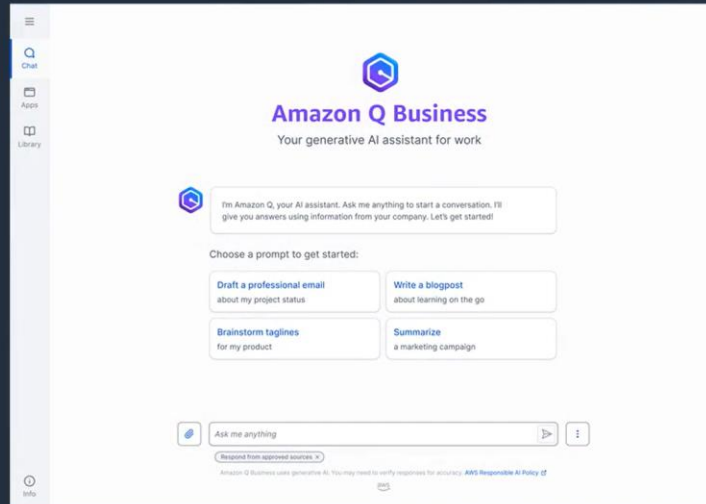
These can be broadly divided into the following categories:



Amazon Q Business

Amazon Q Business

Your generative AI-powered assistant that can answer questions, generate content, create summaries, and complete tasks—all based on the information in your enterprise.



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Common uses for Amazon Q Business



Streamline search experience

Connect internal wikis, SharePoint sites, Confluence, Salesforce, and 40+ other applications for a unified, conversational search experience.



Extract key insights

Perform comparative analysis on your documents and unlock powerful insights so that your employees can make faster, better data-driven decisions.



Generate summaries

Quickly understand the essence of documents by creating summaries of uploaded files or existing enterprise documents.



Accelerate content creation

Empower employees to create content faster and smarter across departments.



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Benefits of Amazon Q Business

Some of the benefits of Amazon Q Business include:

- Quick, accurate, and comprehensive answers
- Simple deployment and management
- Configurable and customizable source options
- Data and application security
- Broad connectivity



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Amazon Q Developer

Amazon Q Developer



Explore new services and capabilities



Streamline research, design, coding, and more



Get quick, accurate answers to questions about AWS and code



Explore how to build secure and stable applications



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Amazon Q is available wherever you interact with AWS:



AWS Management Console



IDEs



AWS Documentation



Slack and Teams (through AWS Chatbot)



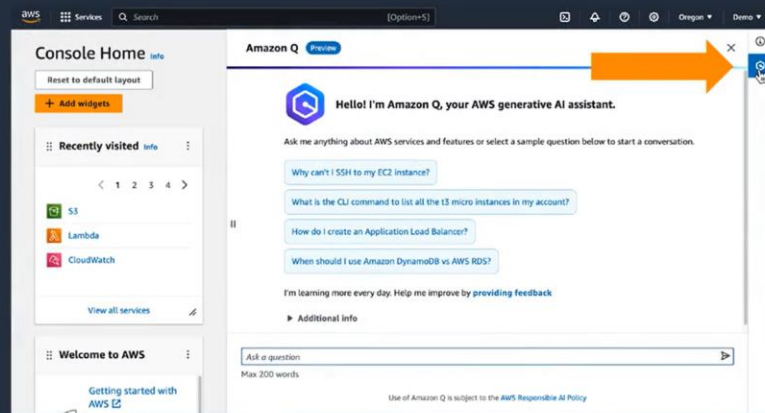
AWS Console Mobile Application



Amazon CodeCatalyst

Amazon Q Developer in the AWS Management Console

Amazon Q Developer is available in the console to provide expert recommendations and step-by-step instructions when getting started with AWS services, implementing the latest best practices, looking for the right service for the job, and more.



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Benefits of Amazon Q Developer in the AWS Management Console

Amazon Q Developer helps you learn, explore, and take action. For example, you can:

- Chat with Amazon Q to explore new AWS capabilities and architect solutions.
- Ask questions about AWS services and features.
- Get guidance on architecture best practices.
- Diagnose and troubleshoot errors and network connectivity issues.
- Connect to AWS support.



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Amazon Q Developer in the IDE

Explore and understand your code, generate tests, build entirely new features, or upgrade your application language version with interactive assistance in the IDE.

```
5 public class SHA256Example {  
6  
7     public static void main(String[] args) {  
8         Scanner sn = new Scanner(System.in);  
9         System.out.print("Please enter the SHA re  
10        String data = sn.nextLine();  
11  
12        SHA256Example shaExample = new SHA256Examp  
13        String hash = shaExample.retrieveSHA256Hash  
14        System.out.println("The SHA256 (hexadecim  
15    }  
16  
17    private String retrieveSHA256Hash(String data)  
18        String result = null;
```

Explain selected code

AMAZON Q

The selected code block is calculating the **SHA-256** hash of a given string. Here's a breakdown:

1. It imports the necessary classes - **MessageDigest** for hashing, **Scanner** for user input, and **DatatypeConverter** for converting bytes to a hexadecimal string.
2. **main()** method takes user input, creates a **SHA256Example** object, and calls **retrieveSHA256Hash()** to get the hash.
3. **retrieveSHA256Hash()** gets the **MessageDigest** for SHA-256 hashing. Hashes the input string bytes and converts to a hex string using **bytesToHex()**.

```
private String retrieveSHA256Hash(String data)
```

Ask a question



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Benefits of Amazon Q Developer in the IDE

Amazon Q Developer can assist in the software development process. It can:

- Help with building on AWS.
- Chat with Amazon Q Developer about software development and code-related questions.
- Generate code for you.
- Update your code language version using Amazon Q Code Transformation.
- Explain, refactor, or optimize code snippets
- Develop features



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Common uses of Amazon Q Developer

Amazon Q Developer can be used across the SDLC :

Plan

- Amazon Q in the console (best practices, EC2 instance optimization, etc.)
- Business-specific application

Create

- In-line coding companion in IDE and CLI
- Feature development
- Chat in IDE

Test & secure

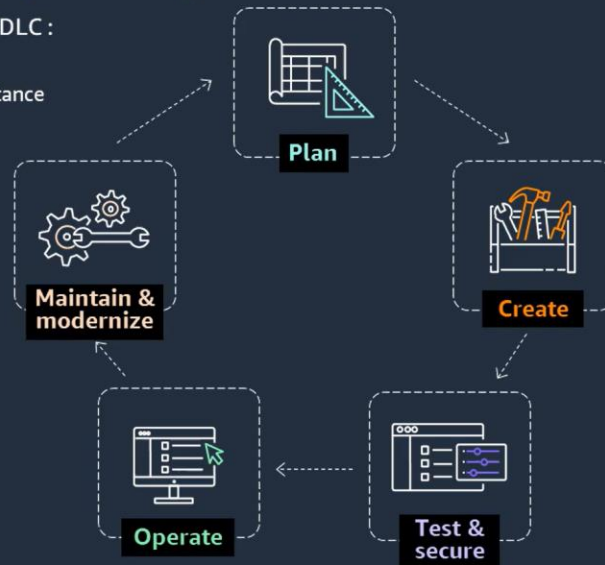
- Unit test generation
- Security scanning & remediation

Operate

- Troubleshoot errors (S3, Lambda, EC2, ECS)
- VPC Reachability Analyzer

Maintain & modernize

- Modernize code with Amazon Q Code Transformation



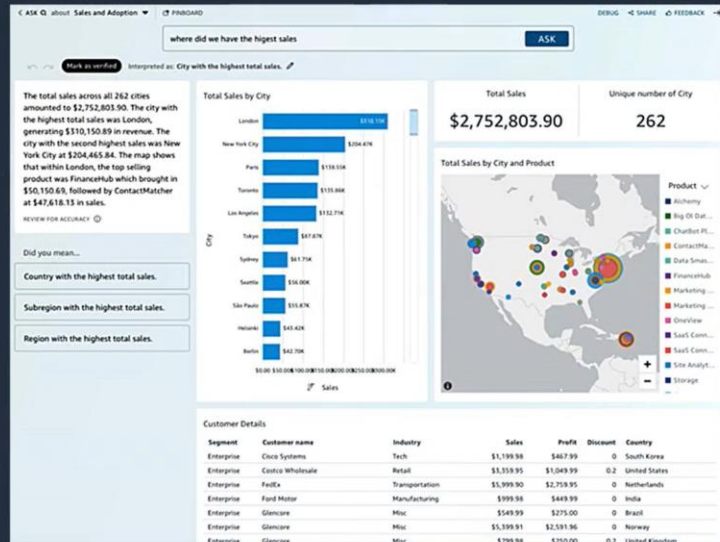
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Amazon Q in QuickSight

Amazon Q in QuickSight

Amazon QuickSight, our machine learning (ML)-powered business intelligence service, now has Amazon Q features built in.

Dashboard authoring experience, insights on demand, and data storytelling are some of its key features.



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Benefits of Amazon Q in QuickSight

The following are some of the benefits of Amazon Q in QuickSight:

- You can use it to reduce time to generate insights.
- Amazon Q is built with security and privacy in mind.
- BI for everyone: Amazon Q makes it simple for everyone to understand data better.

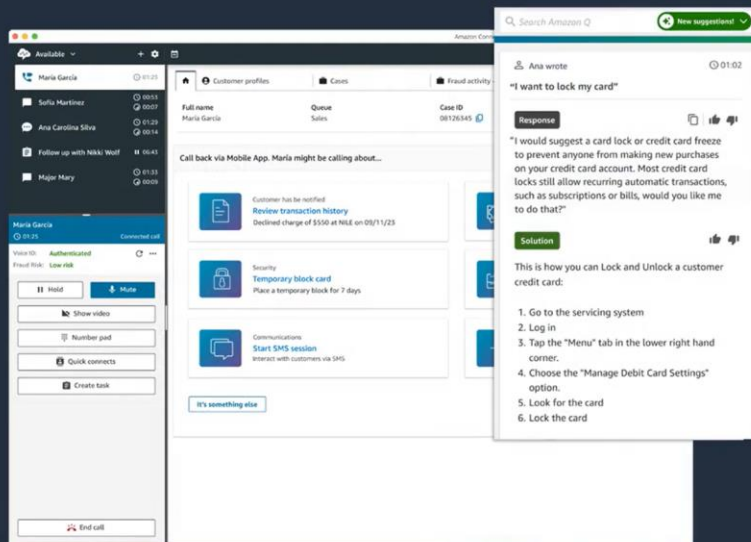


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Amazon Q in Connect

Amazon Q in Connect

In Amazon Connect, our contact center service, Amazon Q helps your customer service agents provide better customer service.



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Amazon Q in AWS Supply Chain

Amazon Q in AWS Supply Chain

Using Amazon Q in Supply Chain (coming soon), inventory managers, supply and demand planners, and others will be able to ask questions about their supply chain.

Amazon Q will provide intelligent answers about what is happening, why it is happening, and what actions to take.

